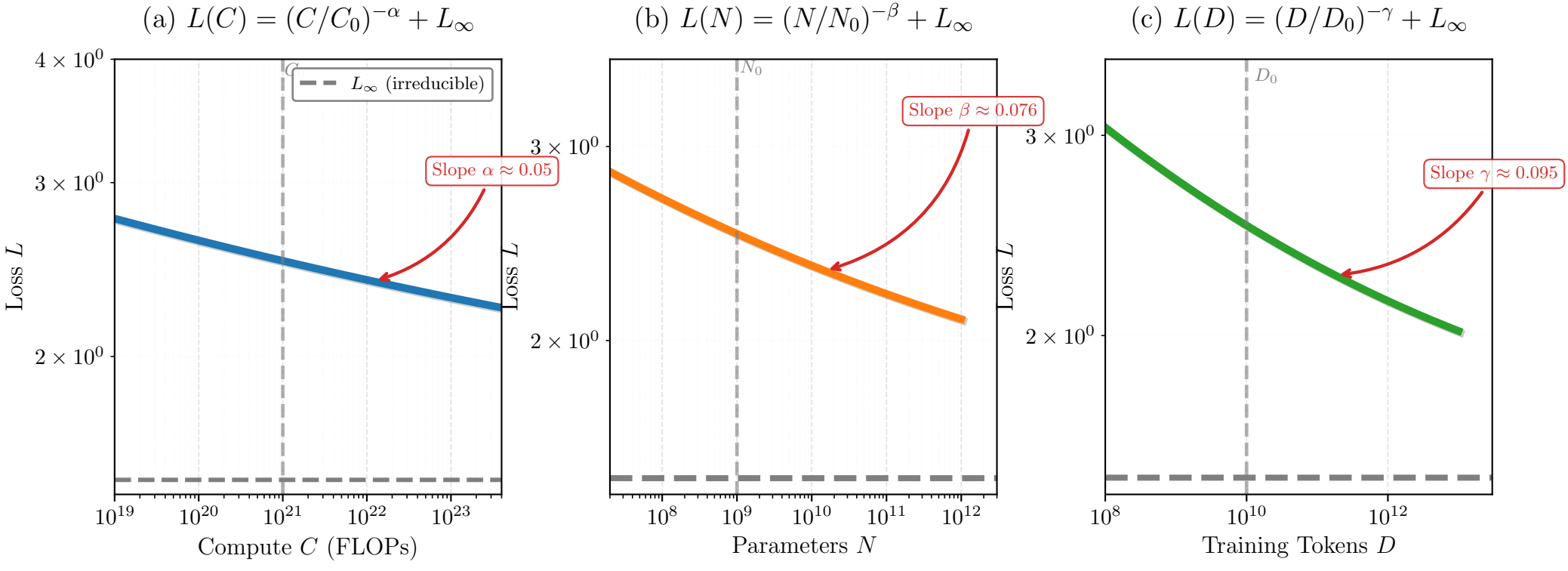
(a) $L(C) = (C/C_0)^{-\alpha} + L_\infty$
(b) $L(N) = (N/N_0)^{-\beta} + L_\infty$
(c) $L(D) = (D/D_0)^{-\gamma} + L_\infty$
L_∞ (irreducible)
Slope $\alpha \approx 0.05$
Slope $\beta \approx 0.076$
Slope $\gamma \approx 0.095$
C_0
N_0
D_0
Loss L
4×10^0
3×10^0
2×10^0
10^{19} 10^{20} 10^{21} 10^{22} 10^{23}
Compute C (FLOPs)
10^8 10^9 10^{10} 10^{11} 10^{12}
Parameters N
10^8 10^{10} 10^{12}
Training Tokens D